Efthymios Papageorgiou

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github.com/epap011 | Google Scholar | Personal Website

Technical Skills

Observability & Load Testing: Prometheus, Grafana, OpenTelemetry, Jaeger, Locust, K6
Distributed Systems & Cloud: Kubernetes, Docker, AWS, Istio/Envoy, gRPC, Ansible, Linux
Programming Languages: C, C++, Go, Python, Bash
Data & Web Servers: Apache Spark, Apache httpd, nginx
Research Areas: Resource Management, Overload Control, Quality of Service, Cloud Computing

Experience

Graduate Research Fellow → Systems Analyst — Mar. 2024 – May. 2026
Foundation for Research and Technology – Hellas (FORTH-ICS) — *Heraklion, Greece*

- Designed a **PPO reinforcement-learning controller** in Kubernetes that dynamically adjusts container CPU shares to control tail latency for containerized microservices. [code]
- Designed an **MPC-based autoscaler** for microservice applications on Kubernetes/Istio that jointly optimizes CPU shares and replica counts under end-to-end response-time SLOs, using a **Layered Queueing Network** performance model; evaluated on DeathStarBench Hotel Reservation.
- Built a **Model Predictive Control** system combining classical and quantum-annealing solvers that held target CPU utilization for an Apache HTTP Server under varying load; published at **IEEE ACSOS 2024**

Research Assistant — DEDALUS Project — Aug. 2025 – Nov. 2025
University of Crete — *Heraklion, Greece*

- Co-developed an end-to-end **quantum-enhanced framework for cost-aware join-order optimization with search-space pruning** for SQL workloads; accepted at **EDBT 2026**.

Undergraduate Research Fellow — Apr. 2023 – Mar. 2024
Foundation for Research and Technology – Hellas (FORTH-ICS) — *Heraklion, Greece*

- Explored the adaptivity mechanisms of the Apache HTTP Server across different **Multi-Processing Modules (MPMs)**

Selected Projects & Publications

QUADS | *ICPE 2026* — Hybrid quantum–classical control of adaptive computer systems.
MPC of Internet Servers via Quantum Annealing | *ACSOS 2024* — Real-time CPU utilization control for Apache HTTP Server combining classical MPC with quantum solvers.
DEDALUS | *EDBT 2026* — Quantum-enhanced end-to-end framework for cost-aware join-order optimization with search-space pruning.
Cardinality Estimation for Query Optimization | *Future of DB Programming Contest, ATHENA RC / Huawei, 2024* — Ranked **12 / 60** on a cardinality-estimation task.

Honors & Awards

2025 — IPTO (ADMIE) Scholarship, Independent Power Transmission Operator of Greece
2024 — Graduate Scholarship, Foundation for Research and Technology – Hellas (FORTH)
2024 — EuroSys 2024 Student Grant Recipient
2023 — Undergraduate Scholarship, Foundation for Research and Technology – Hellas (FORTH)
2023, 2025 — ICPC Greece Regional (ranked 41/87), advanced to ICPC Europe phase

Education

University of Crete — Heraklion, Greece
Ph.D. in Computer Science & Engineering — *Sep. 2025 – Present*

University of Crete — Heraklion, Greece
M.Sc. in Computer Science & Engineering (GPA: 9.21/10, Thesis: 10/10) — *Sep. 2023 – Jul. 2025*

University of Crete — Heraklion, Greece
B.Sc. in Computer Science (GPA: 7.61/10, Thesis: 10/10) — *Sep. 2019 – Sep. 2023*